Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam E · Form 2: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. Object A, whose mass is 14 times B's, collides with B. Ignore other horizontal forces during contact. Which comparison of force magnitudes and acceleration magnitudes is correct?

(A) Forces are equal in magnitude; B's acceleration magnitude is 14 times A's.

(B) A exerts 14 times the force; accelerations are equal.

(C) Forces are equal; A's acceleration is 14 times B's.

(D) Different masses make both comparisons impossible.

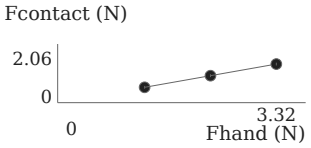
2. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Push inward on the 3 kg cart to drive the 2 kg cart. Compare force magnitudes. What is the new intercept minus the original intercept?

(A) 0

(B) 0.126

(C) -0.252

(D) 0.378



3. A 40 kg student stands on a scale in an elevator that is moving up but slowing down, with acceleration magnitude 3 m/s². What force does the scale read? Use g = 9.8 m/s².

(A) 544 N

(B) 136 N

(C) 272 N

(D) -272 N

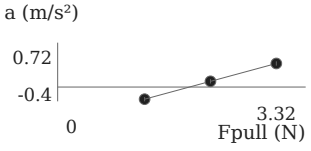
4. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.32, -0.34); (2.32, 0.16); (3.32, 0.66). With vertical error bounded by 0.06 and an exact model, what is the error bound on an inferred X?

(A) 0.12

(B) 0.37

(C) 0.245

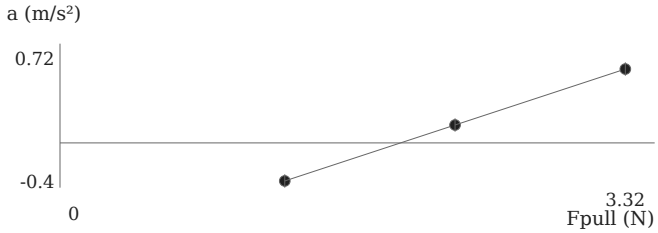
(D) -0.005



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.32, -0.34); (2.32, 0.16); (3.32, 0.66).

a. For the cart under tow setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double cart mass while holding external resistance at 2 N. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.32, 566); (2.32, 616); (3.32, 666).

a. Build the model for passenger on a scale from physics and predict both slope and intercept. **(2 pt)**

b. Double passenger mass in the same gravity. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1.32, 0.792); (2.32, 1.392); (3.32, 1.992). Build the model for two carts in contact from physics and predict both slope and intercept. **(3 pt)**